PROJECT REPORT

Driver Drowsiness and Head Movement Detection System Using AI

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Introduction

Driver fatigue and distraction are major causes of road accidents. Drowsiness can cause slow reactions, while abnormal head movements (like nodding or tilting) can indicate loss of attention or sleep.

This project presents an **AI-based Driver Monitoring System** that detects both **eye closure and head movement** using a webcam. If signs of drowsiness or unsafe head posture are detected, the system alerts the driver to prevent accidents.

Problem Statement

Road accidents due to sleep and lack of attention are increasing. Most existing systems:

- Focus only on eye detection
- Ignore head movement patterns
- Are expensive and complex

This project solves the problem by creating a **low-cost system that detects both eye closure and abnormal head movement** in real time.

Objectives

- To detect eye closure using a webcam
- To monitor head movement (nodding, tilting)
- To identify drowsiness and distraction
- To alert the driver immediately
- To improve road safety

Proposed Solution

The system works by:

- Capturing live video from a webcam

- Detecting face landmarks
- Tracking eye status (open/closed)
- Tracking head position and movement
- Triggering an alert if:
    - Eyes are closed for too long
    - Head tilts or nods abnormally

Methodology

1. Video Capture

- Continuous webcam input

2. Face Detection

- Detects face in each frame

3. Facial Landmark Detection

- Identifies key points (eyes, nose, jaw)

4. Eye Closure Detection

- Calculates Eye Aspect Ratio (EAR)
- Determines if eyes are closed

5. Head Movement Detection

- Tracks head tilt and position
- Detects:
    - Nodding (sleep sign)
    - Side tilting (distraction)

6. Decision Logic

- If EAR is below threshold → eyes closed
- If head angle exceeds limit → unsafe movement

7. Alert System

- Alarm sound or warning message

System Architecture

Webcam → Face Detection → Landmark Detection → Eye + Head Analysis → Alert System

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Implementation

Tools & Technologies Used:

- Python 3.13
- VS Code

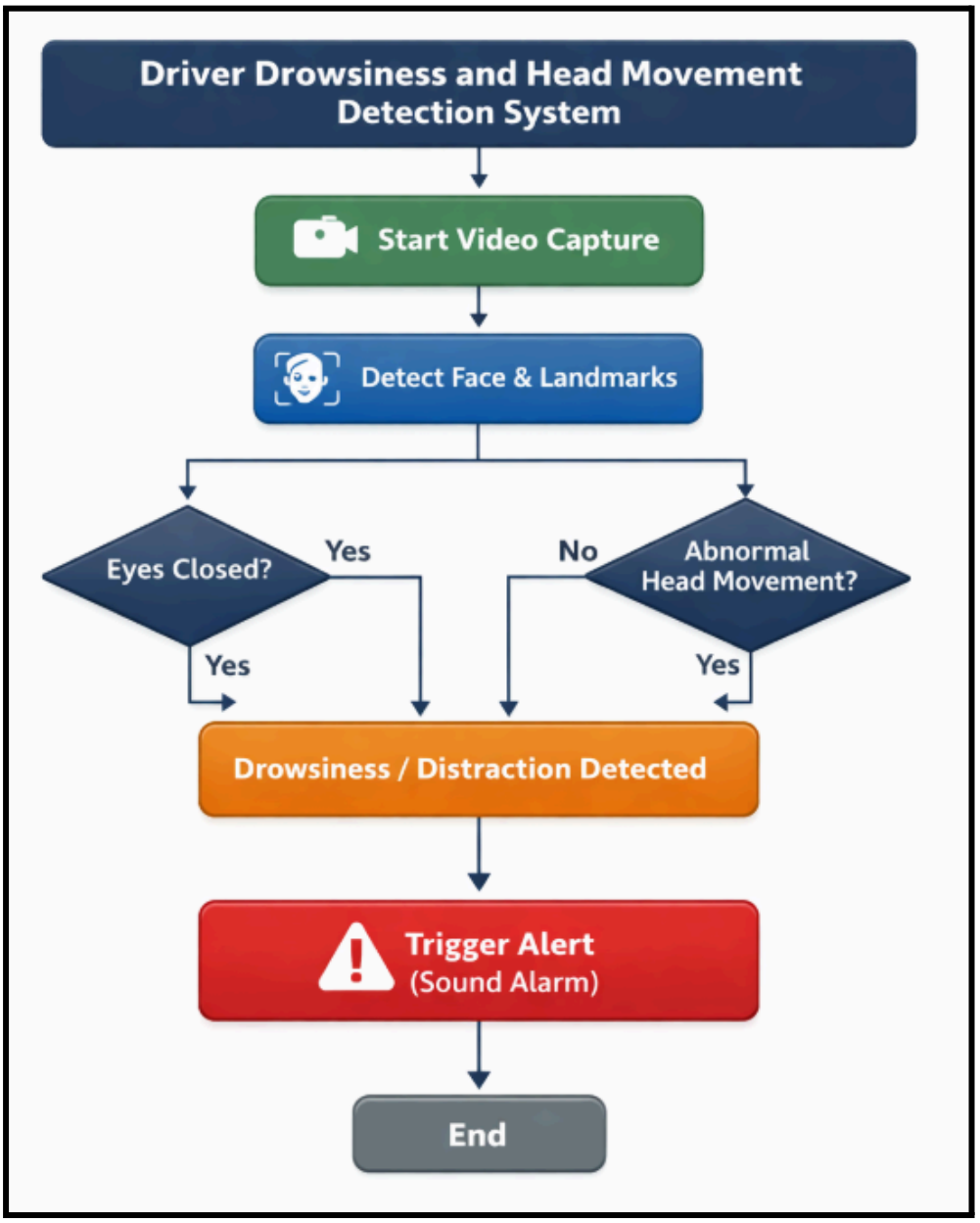
Libraries:

- OpenCV
- imutils
- scipy

Key Features

- Real-time drowsiness detection
- Head movement tracking
- Dual safety system (eyes + head)
- Instant alert mechanism
- Cost-effective solution

Flowchart



Key Decisions

- Combined eye detection with head tracking for better accuracy
- Used facial landmarks for precise monitoring
- Implemented threshold-based logic for simplicity

Challenges Faced

- Detecting head angles accurately
- Differentiating normal movement vs drowsiness
- Handling lighting variations
- Reducing false alerts

Results

The system successfully:

- Detected eye closure in real-time
- Identified abnormal head movements
- Triggered alerts effectively
- Improved overall detection accuracy compared to eye-only systems

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Learning Outcomes

- Learned facial landmark detection
- Understood real-time computer vision systems
- Improved Python and debugging skills
- Learned how to combine multiple features for better AI performance

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Future Improvements

- Add deep learning models for higher accuracy
- Detect yawning and facial expressions
- Work in night/low-light conditions
- Integrate with vehicle hardware
- Add mobile app notifications

Conclusion

This project demonstrates an effective AI-based system to prevent accidents caused by drowsiness and distraction. By combining **eye closure detection and head movement tracking**, the system provides a more reliable safety mechanism and has strong potential for real-world applications.